Quantitative Risk Analytics Report

Small Controlled Expansion After Pilot 01 Stress Maps

Codex-assisted diagnostic analytics report

June 17, 2026

Abstract

This report documents a small quantitative risk-analytics expansion after Pilot 01. The experiment uses the validated baseline American Crank–Nicolson / projected SOR solver only. It quantifies boundary movement, continuation premium, Gamma concentration, PSOR effort, runtime, and LCP diagnostics under selected one-at-a-time stress parameters. The outputs are diagnostic analytics only. They are not broad stress-map datasets, training labels, neural-surrogate inputs, or production risk surfaces.

Contents

1 Purpose of Quantitative Risk Analytics

Pilot 01 showed that the validated baseline solver can produce small, metadata-rich diagnostic stress-map figures. The present step is narrower than data generation but more quantitative than visual inspection. Its purpose is to summarize how key numerical and financial diagnostics move when selected stress parameters are changed one at a time.

The experiment follows the downstream design gate and Ticket 12 synthesis decision, `PASS_SOLVER_VALIDATION_WITH_LIMITATIONS`. It keeps the same classical finite-difference American option setting described by the project validation ladder (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Link to Pilot 01 and Senior Handout Experiment 04

The senior handout's risk-analytics stage asks for boundary movement, Gamma concentration, PSOR iteration effort, runtime, and complementarity diagnostics before dataset construction. This report implements that idea at small scale. It deliberately does not export training arrays or machine-learning-ready targets.

Handout theme	This report's controlled implementation
Boundary movement	Selected boundary values and boundary curves across stress sweeps.
Gamma concentration	Full-grid and strict-mask maximum $\lvert\Gamma\rvert$ summaries.
PSOR effort	Mean and maximum PSOR iterations plus runtime seconds.
Complementarity diagnostics	Obstacle, equation, and complementarity product metrics.
Dataset caution	Outputs are marked `analytics\_diagnostic\_only`.

3 Solver and Grid Choice

All runs use the Ticket 04 baseline American CN/PSOR solver:

```
american\_crank\_nicolson\_psor\_price
```

Rannacher smoothing is not used. Ticket 10A kept Rannacher as an optional comparison tool only, so this analytics step does not silently change the validated solver.

Every run uses normalized strike $K = 1, T = 1, S_{\text{max}} = 4$, and $M = N = 120$. The main interpretation region remains $S/K \in [0.4, 1.8]$, while the solver still computes on the full domain.

4 Stress Parameter Design

Family	Sweep	Values
American put	Volatility	$\sigma \in \{0.20, 0.40, 0.60\}$, with $r = 0.05$, $q = 0.02$.
American put	Dividend yield	$q \in \{0.00, 0.02, 0.04\}$, with $r = 0.05$, $\sigma = 0.20$.
American put	Interest rate	$r \in \{0.03, 0.05, 0.08\}$, with $q = 0.02$, $\sigma = 0.20$.
Dividend call	Dividend yield	$q \in \{0.03, 0.08, 0.10, 0.14\}$, with $r = 0.05$, $\sigma = 0.20$.
Dividend call	Volatility	$\sigma \in \{0.20, 0.40, 0.60\}$, with $r = 0.05$, $q = 0.08$.
Dividend call	Interest rate	$r \in \{0.03, 0.05, 0.08\}$, with $q = 0.08$, $\sigma = 0.20$.
Dividend call	Small q-σ diagnostic	$q \in \{0.03, 0.08, 0.10, 0.14\}$ and $\sigma \in \{0.20, 0.40, 0.60\}$.

The experiment writes 31 analytics rows: 19 one-at-a-time sweep rows and 12 small q-σ diagnostic heatmap rows. Duplicate parameter solves are cached internally, but the CSV rows keep their sweep membership.

5 Metric Definitions

Metric	Meaning
Boundary moneyness	Threshold-based boundary spot divided by K, extracted from $U - \Phi$.
Continuation premium	$U - \Phi$, the value above immediate exercise payoff.
PSOR effort	Maximum and mean projected-SOR iteration counts over time steps.
Runtime seconds	Wall-clock runtime for the baseline solver and attached diagnostics.
Obstacle violation	$\max(\Phi - U, 0)$.
Equation violation	$\max(b - AU, 0)$ from the reconstructed LCP.
Complementarity product	$(U - \Phi)(AU - b)$, summarized by maximum absolute value.
Full max $\lvert\Gamma\rvert$	Maximum absolute finite-difference Gamma over finite grid entries.
Strict max $\lvert\Gamma\rvert$	Maximum absolute Gamma after excluding boundary-near, payoff-kink, and maturity rows.

6 Tables and Interpretation

6.1 Boundary Direction Checks

The main qualitative handout directions are visible in the generated numbers. At $\tau/T = 1$, the American put boundary falls as volatility rises:

$$0.7668 \rightarrow 0.5667 \rightarrow 0.4000$$

for $\sigma = 0.20, 0.40, 0.60$. This matches the expected direction: higher volatility raises the value of waiting, so the put exercise boundary moves lower.

For the dividend-paying call, the boundary also falls as dividend yield rises:

$$1.8979 \to 1.2666 \to 1.2332 \to 1.1666$$

for $q = 0.03, 0.08, 0.10, 0.14$. This matches the expected direction that higher dividend yield makes early exercise more attractive for calls.

Sweep	Boundary at $\tau/T = 1$	Interpretation
Put $\sigma = 0.20, 0.40, 0.60$	0.7668, 0.5667, 0.4000.	Decreasing; matches expected volatility direction.
Call $q = 0.03, 0.08, 0.10, 0.14$	1.8979, 1.2666, 1.2332, 1.1666.	Decreasing; matches expected dividend-yield direction.
Call $\sigma = 0.20, 0.40, 0.60$	1.2666, 1.7331, 2.3660.	Increasing; higher volatility delays high-spot exercise in this diagnostic.

6.2 PSOR, Runtime, and LCP Checks

All 31 analytics rows converged and were marked acceptable for analytics review. The largest equation violation was 4.8938×10^{-8}, and the largest complementarity product was 4.7325×10^{-10}. These are small diagnostic values, but they are not a mathematical proof across all future parameter settings.

Volatility increased PSOR effort and runtime in both families. For the put volatility sweep, mean PSOR iterations rose from 6.99 to 29.42 as σ moved from 0.20 to 0.60. The runtime for the same sweep rose from about 0.049 seconds to about 0.204 seconds. For the call volatility sweep, mean iterations rose from 7.35 to 14.92.

Diagnostic	Minimum / baseline reading	Maximum observed	Comment
Obstacle violation	0	0	Payoff obstacle respected in all runs.
Equation violation	Near 10^{-9} for base cases	4.8938×10^{-8}	Largest value occurs in high-volatility put run.
Complementarity product	Around 10^{-11} for many runs	4.7325×10^{-10}	Still small relative to the diagnostic threshold.
Mean PSOR iterations	Around 7 in base cases	29.42	High volatility increases iteration effort.
Runtime seconds	Around 0.05 in base cases	0.204	Small absolute runtime in this pilot.

6.3 Gamma Concentration

Full-grid maximum $|\Gamma|$ is 30 across the runs because the full metric includes fragile regions such as payoff kink and maturity-row effects. The strict-mask maximum is much smaller, around 1.92 to 3.29

in the one-at-a-time sweeps. This difference is exactly why the project keeps Gamma diagnostic and masked. The strict-mask Gamma values do not move monotonically in every sweep, so the report does not claim a simple universal Gamma direction.

7 Figures and Interpretation

Figure 1 shows the scalar put boundary direction check. It is the cleanest evidence that the selected put boundary decreases with volatility in this pilot.

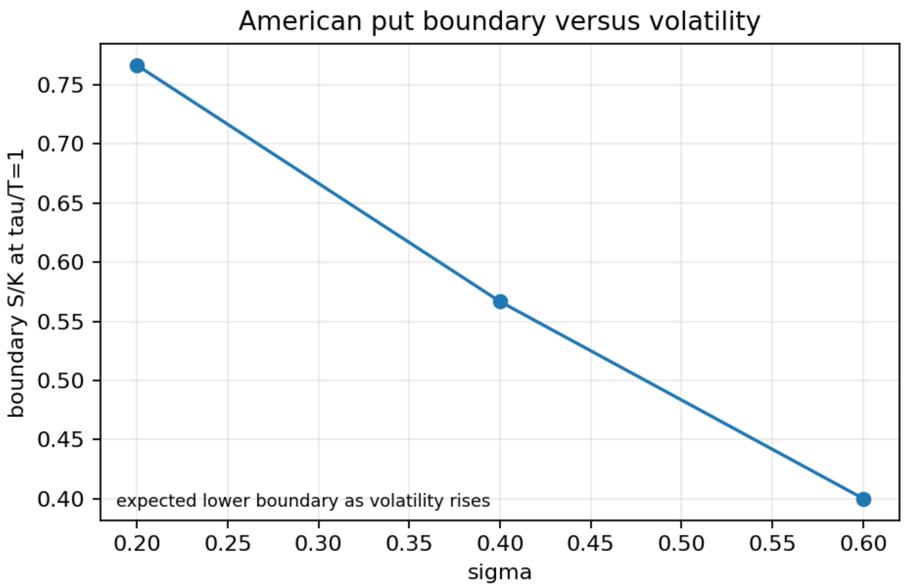

Figure 1: American put boundary moneyness versus volatility at $\tau/T = 1$.

Figure 2 shows the corresponding dividend-call direction check. The boundary decreases as dividend yield rises.

Figures 3 and 4 show the whole threshold-extracted boundary curves over time-to-maturity. They are approximate diagnostic curves, not exact analytical free-boundary solutions.

Figure 5 separates full-grid Gamma from strict-mask Gamma. The full values are dominated by fragile locations; the strict values are the safer summary for human review, but still not production Greek labels.

Figure 6 shows numerical workload. The high-volatility cases are visibly more expensive in iteration count and runtime, matching the senior handout's warning that stress regimes can increase solver effort.

Figure 7 shows LCP diagnostic magnitudes. The values remain small, with the largest equation violations appearing in higher-volatility runs.

Figure 8 is the optional small dividend-call q-σ boundary heatmap. It is an aggregate diagnostic visualization only. It is not a saved data array and not a dataset construction step.

8 Qualitative Comparison with Senior Handout

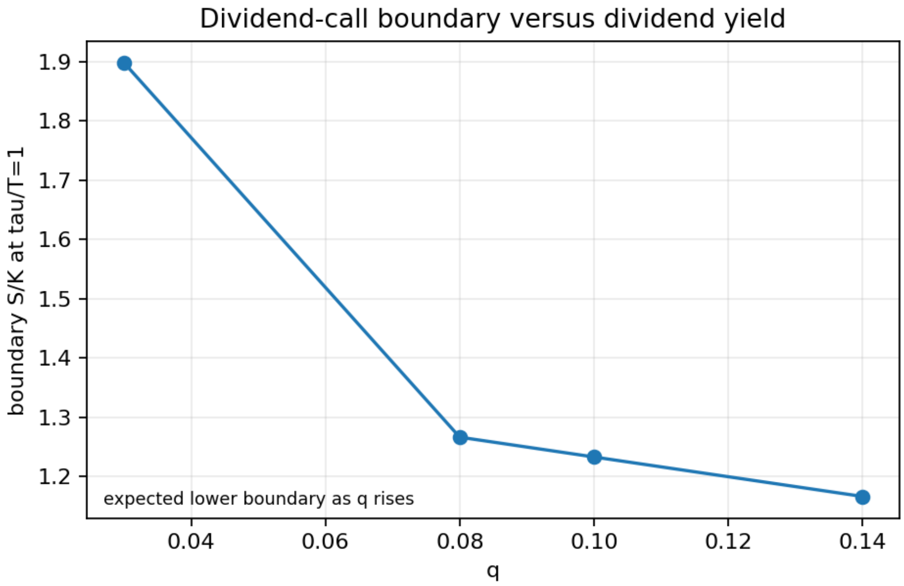

Figure 2: Dividend-call boundary moneyness versus dividend yield at $\tau/T = 1$.

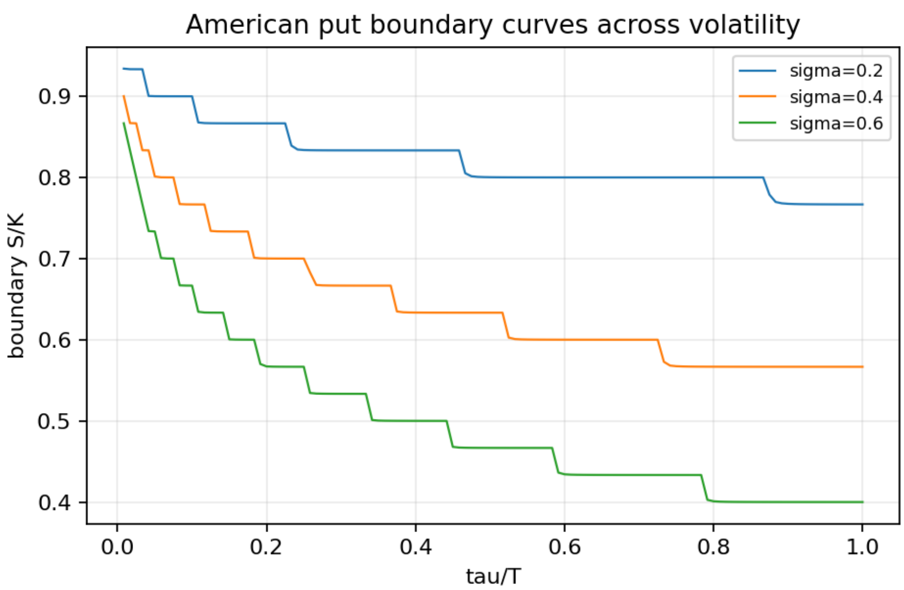

Figure 3: American put boundary curves across the volatility sweep.

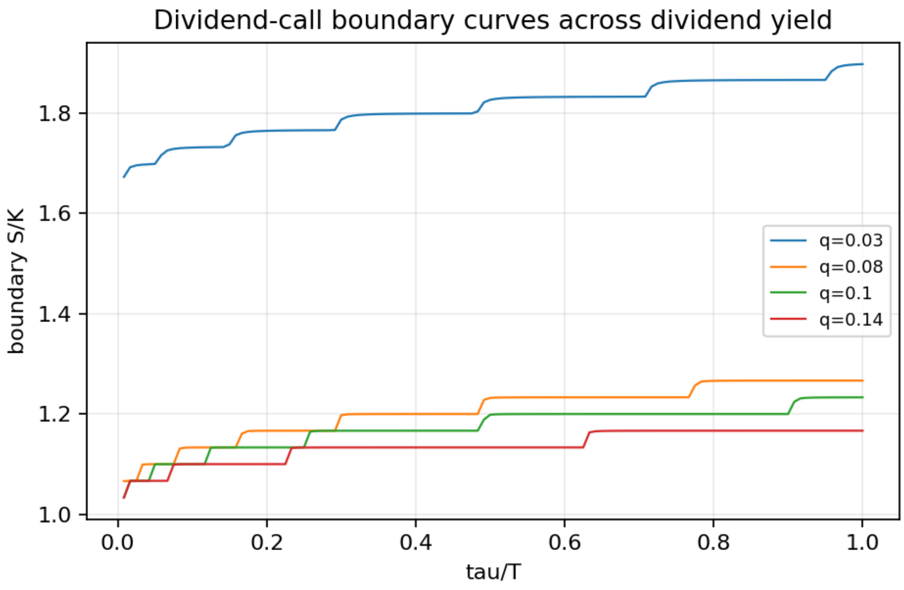

Figure 4: Dividend-call boundary curves across the dividend-yield sweep.

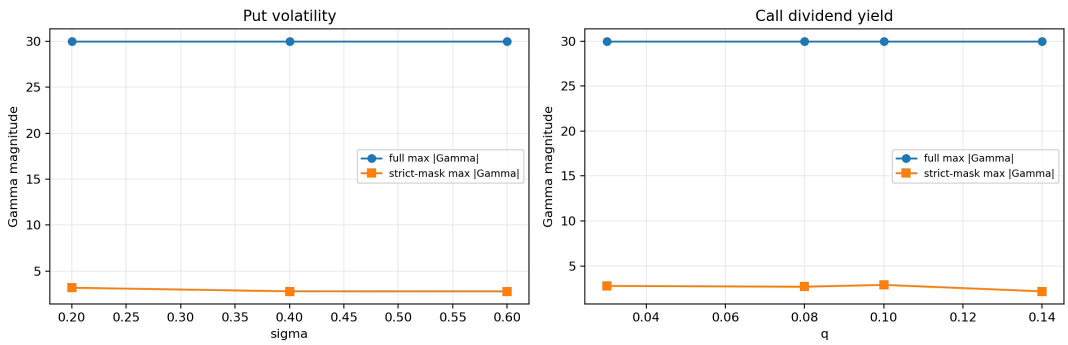

Figure 5: Gamma concentration across key stress sweeps.

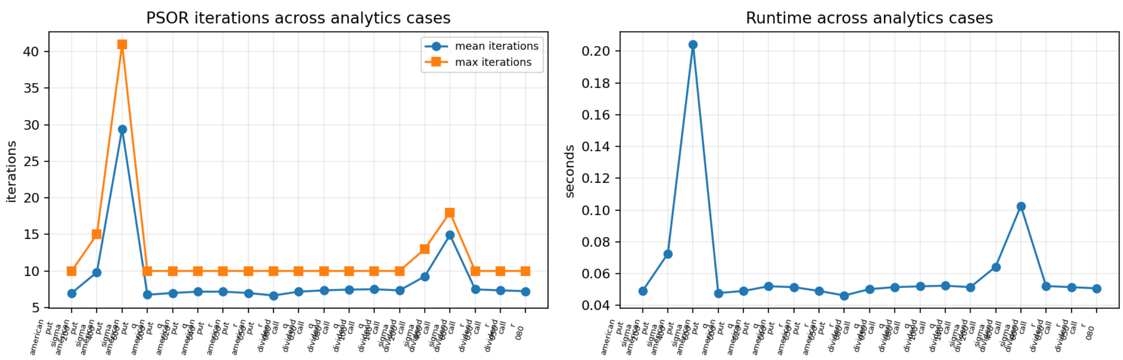

Figure 6: PSOR iteration and runtime diagnostics across analytics cases.

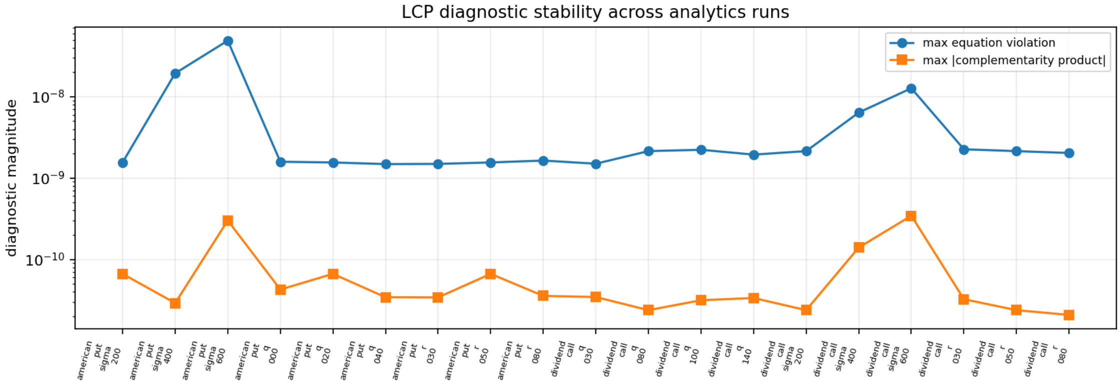

Figure 7: LCP diagnostic stability across analytics cases.

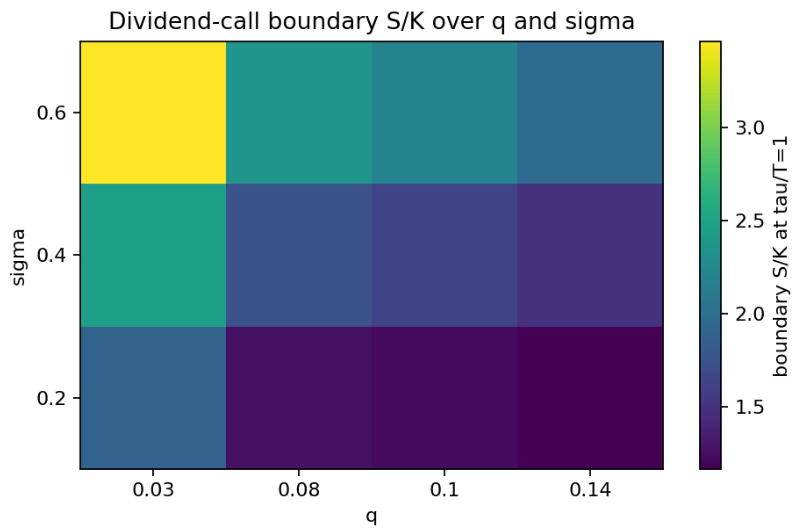

Figure 8: Small aggregate dividend-call boundary moneyness heatmap over q and σ.

Expected direction or caution	Result	Comment
Put boundary should decrease as volatility rises.	Matches.	Boundary at $\tau/T = 1$ falls from 0.7668 to 0.4000.
Dividend-call boundary should decrease as dividend yield rises.	Matches.	Boundary falls from 1.8979 at $q = 0.03$ to 1.1666 at $q = 0.14$.
Higher volatility may increase PSOR effort and runtime.	Matches.	Put mean iterations rise to 29.42; runtime rises to 0.204 seconds.
Gamma concentration should be strongest near kink/boundary regions.	Caution confirmed.	Full max $\lvert\Gamma\rvert$ is dominated by fragile full-grid locations; strict-mask values are lower.

9 Acceptance Checks

Criterion	Result	Interpretation
Use baseline CN/PSOR solver only	PASS	No Rannacher run is used.
Keep sweeps small and controlled	PASS	19 one-at-a-time rows plus 12 small heatmap rows.
All PSOR steps converge or are reported	PASS	All rows converged.
Obstacle violation near zero	PASS	Maximum obstacle violation is 0.
LCP diagnostics near tolerance	PASS	Diagnostics remain small in all rows.
Boundary metadata present	PASS	Selected-tau boundary rows include found/not-found status.
Greek diagnostics masked and cautious	PASS	Greek table includes strict-mask and caution metrics.
No dataset or neural outputs	PASS	Only CSV summaries, PNG figures, and this report are created.

10 Limitations

- The q-σ heatmap is a small aggregate diagnostic, not a broad two-dimensional parameter sweep.
- Boundary extraction is threshold-based with threshold 10^{-6}.
- The scalar direction checks use the nearest $\tau/T = 1$ row and should not be read as exact comparative statics.
- Gamma remains finite-difference diagnostic output. It is not a production Greek measure.
- Runtime seconds are local-machine observations, useful for this pilot but not a portable performance benchmark.
- The experiment supports human review, not automated downstream generation.

11 What This Analytics Step Supports

This step supports the claim that the validated baseline solver can produce small quantitative diagnostic analytics aligned with the senior handout's risk-analytics expectations. It also supports the specific qualitative observations that the put boundary decreases with volatility, the dividend-call boundary decreases with dividend yield, and high volatility increases PSOR effort in the tested settings.

12 What This Analytics Step Does Not Support

This step does not support broad stress-map generation, exact free-boundary claims, production Greek reporting, training-label export, neural-surrogate readiness, or final paper-style claims about all parameter regimes. Those remain blocked until a separate human-reviewed plan approves the next stage.

13 Recommended Next Step

The recommended next step is human review of the analytics tables, figures, and limitations. If accepted, the next technical step should be another small approved pilot that either enlarges the stress grid modestly or compares selected analytics at a higher grid resolution. It should still avoid dataset construction and surrogate-model training.

14 Reproducibility Record

Item	Record
Source module	`src/american_risk_surfaces/downstream/risk_analytics.py`.
Experiment script	`experiments/12_quantitative_risk_analytics.py`.
Test file	`tests/test_quantitative_risk_analytics.py`.
Output folder	`results/03_quantitative_risk_analytics/`.
Report source	`reports/04_downstream/quantitative_risk_analytics_report.tex`.
Report PDF	`reports/04_downstream/quantitative_risk_analytics_report.pdf`.
Bibliography	`reports/03_solver/references.bib`.

References

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